

RSSI-Based Passenger Movement Classification

Non-Intrusive Public Transport Monitoring

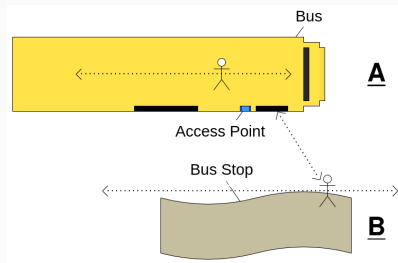
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A Smarter Public Transport

The **inMotion** project aims to bring real-time intelligence to public transport operations. One of its core goals: understand **how passengers move** through the network: who boards, who alights, and where. Today we present a component of that vision: passenger movement classification using only Wi-Fi signals.



Experimental prototype at Instituto de Telecomunicações

What operators need:

- Accurate passenger flow data for scheduling, fleet sizing, and route planning
- Coverage across the entire fleet, not just a few instrumented vehicles
- Low deployment and maintenance cost

Four Movements to Classify

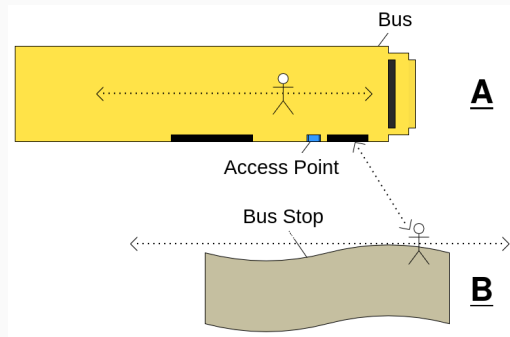
We want to answer a simple question for every passenger interacting with the bus door:

Did this person board, alight, or stay put?

Label	Meaning	Description
AA	$A \rightarrow A$	Remaining inside the bus
BB	$B \rightarrow B$	Remaining at the bus stop
BA	$B \rightarrow A$	Boarding the bus
AB	$A \rightarrow B$	Alighting the bus

Zone A = bus interior (near access point)

Zone B = bus stop (outside)



Two-zone environment: door separates A from B

Using Wi-Fi Signals as Movement Signatures

One access point. Ten seconds of RSSI. Four movement classes.

No specialized hardware

Standard Wi-Fi AP already on
many buses

No device identification

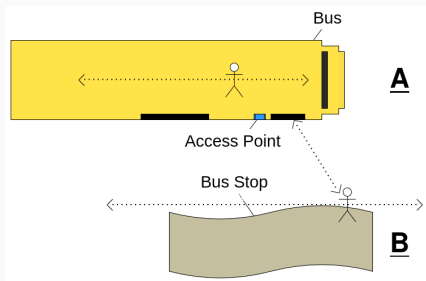
We track signal patterns, not
people

No active participation

Uses existing Wi-Fi
associations passively

The temporal evolution of RSSI encodes whether a device is moving toward or away from the access point.

Experimental Setup



MAYBE CHANGE HERE TO A REAL PHOTO??

Controlled indoor environment at IT Aveiro

Physical setup:

- Zone A: closed room (bus interior), AP at doorway
- Zone B: corridor (bus stop), wall attenuation between zones

Collection:

- 10 s windows at 1 Hz $\rightarrow$ 10 RSSI values per sample
- 4 smartphones, 3 brands
- Scripted movements with ground truth labels
- MAC-based device isolation for clean trajectories

What We Collected

1,356 labelled samples across two conditions:

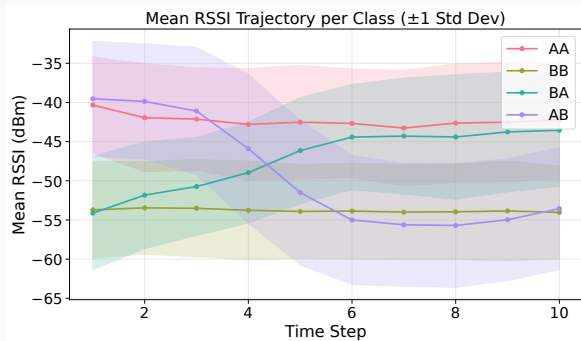
Condition	Samples	Description
Isolated	160	One device at a time, clean signals
Noisy	1,196	Four simultaneous devices, interference

Approximately 340 samples per movement class. The dataset is publicly available on IEEE Dataport and Zenodo.

ID	t1	t2	t3	t4	t5	t6	t7	t8	t9	t10	Class
1	-42	-44	-48	-52	-55	-58	-60	-62	-63	-64	AB
2	-65	-63	-58	-54	-50	-47	-45	-43	-42	-41	BA
3	-40	-41	-39	-42	-40	-41	-39	-40	-41	-40	AA
4	-62	-63	-61	-64	-62	-63	-61	-62	-63	-62	BB

Four samples from the dataset: each row is a 10-second RSSI trajectory

Temporal RSSI Trajectories



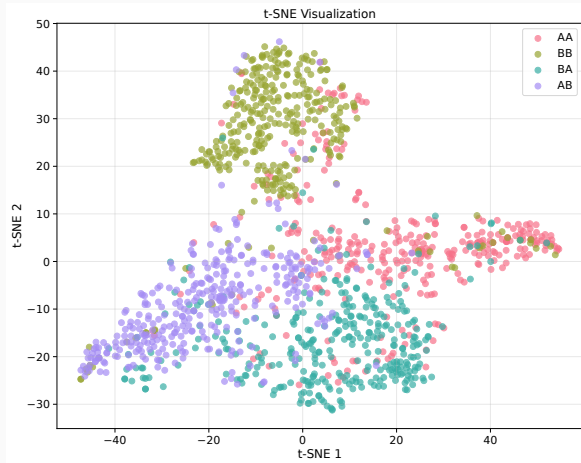
Mean RSSI trajectory per class over 10 seconds (± 1 std)

Clear temporal signatures emerge:

- **Boarding (BA):** steady RSSI increase as the device approaches the AP
- **Alighting (AB):** pronounced RSSI decrease as the device moves away
- **Static states (AA, BB):** stable signal at distinct levels: AA higher (near AP), BB lower (outside)

These contrasting trends are the foundation of the classification.

t-SNE Visualization



t-SNE projection of the 10-dimensional RSSI feature space

Classes form well-defined clusters

confirming that the collected samples carry structure for classification.

Overlap between AA and BA expected: both involve strong RSSI near the AP, but differ in temporal pattern.

38 classifiers from six families:

- SVMs (RBF, Linear)
- Ensemble methods (Random Forest, Extra Trees, CatBoost, XGBoost, LightGBM)
- Gaussian Process
- Neural Networks (MLPs, small to large)
- Regularized logistic regression (L1, L2, ElasticNet)
- Stacking and Voting ensembles

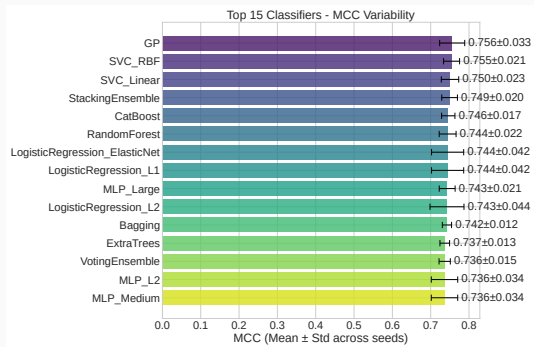
Rigorous evaluation protocol:

- Stratified 80/20 train-test split
- 5-fold stratified cross-validation
- 3 random seeds for stability
- Bayesian hyperparameter optimization (Optuna, 50–1300+ trials per classifier)

Primary metric: Matthews Correlation Coefficient (MCC)

It produces high scores only when all four quadrants of the confusion matrix perform well.

MCC Comparison

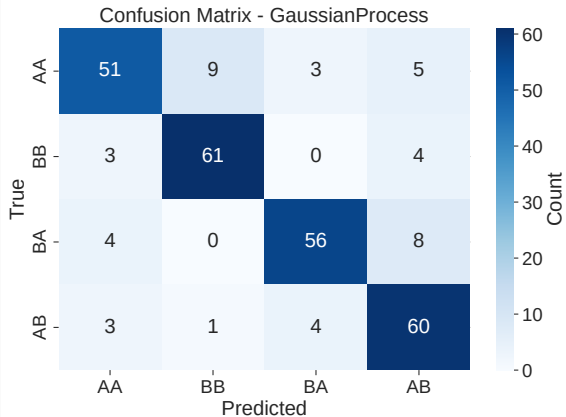


MCC variability across three random seeds (combined dataset)

Classifier	Combined	Isolated	Noisy
Gaussian Process	0.756	0.414	0.755
SVC (RBF)	0.755	0.825	0.754
CatBoost	0.746	0.782	0.770
KNN (k=5)	0.692	0.907	0.704

Mean MCC ± std across 3 seeds

Gaussian Process on Combined Dataset



Normalized confusion matrix, Gaussian Process

Class	Precision	Recall	F1
AA (Inside)	0.78	0.73	0.75
BB (Stop)	0.87	0.89	0.88
BA (Boarding)	0.83	0.77	0.80
AB (Alighting)	0.78	0.88	0.83
Weighted Avg	0.82	0.82	0.82

Strong diagonal dominance. Errors are concentrated between spatially adjacent classes: AA/BA and AB/BB.

The Effect of Interference

Scenario	Best Classifier	MCC	AA F1	BB F1	BA F1	AB F1
Isolated	KNN (k=5)	0.907	0.87	0.96	0.91	0.96
Noisy	CatBoost	0.770	0.79	0.87	0.82	0.83
Combined	Gaussian Proc.	0.756	0.75	0.88	0.80	0.83

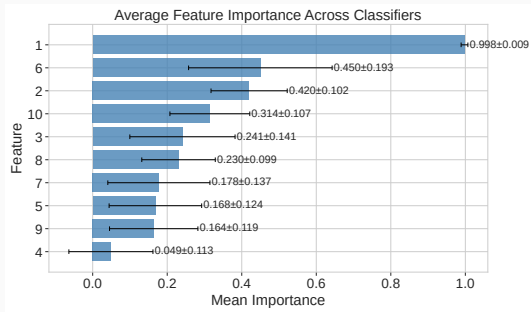
Clean signals: near-perfect

F1 scores above 0.86 for every class. Simple classifiers (KNN) perform best. Upper bound on what RSSI can achieve.

Realistic conditions: robust

Per-class F1 above 0.77 with interference.
Ensemble methods handle noise best.

What the Classifiers Actually Use



Mean feature importance across interpretable classifiers (min-max normalized)

- **Feature 1:** initial RSSI, universal classifier agreement
- **Feature 6:** mid-trajectory, confirms movement direction
- **Feature 2:** early trajectory, reinforces starting position
- **Feature 10:** end of window, confirms movement direction

Sequential measurements outperform aggregate statistics. The temporal order matters.

What we have demonstrated:

- **Four classes separable** from 10 RSSI values
- **MCC 0.907** in clean conditions, **0.770** with interference
- **Single access point** at the door, no extra hardware
- **Privacy-preserving**: no device identification, no localization

What we must be careful about:

- **Controlled environment**: real buses add vibration, crowding, body absorption
- **10 s window** may not fit all movement speeds
- **Only associated devices** are visible; Wi-Fi-off passengers go undetected

This work:

- **Temporal RSSI classification** of 4 movement patterns
- **38 classifiers** evaluated with Bayesian optimization
- **MCC 0.756** combined, **0.907** clean, **0.770** noisy

The dataset is public.

IEEE Dataport · Zenodo · GitHub

Next steps:

- **Operational validation:** real bus deployment
- **Adaptive windows:** dynamic length based on movement speed
- **Sensor fusion:** combine with other modalities
- **OD matrix estimation:** aggregate events across stops
- **Noise robustness:** close the 0.907 → 0.770 gap

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Thank you. Questions?

Dataset: IEEE Dataport (10.21227/55nm-0r91) · Code: github.com/ATNoG/inMotion